

Sharpe Ratio Maximisation:^{*} a convex reformulation

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I wrote this originally for a homework question but I find it cool enough (5°) to make public. I might add some pictures to make it cooler (2°).

Setup

Fix the number of risky assets $n \geq 1$; we are given a set of whose elements are called *feasible portfolios* (I think this could be relaxed to any convex set, but that's how it was given in the problem)

$$X = \{x \in \mathbb{R}^n : f_i(x) \leq \tau_i, i = 1, \dots, m, Ax \leq b\}$$

for some positive-homogenous subadditive f_i s and a bounded polyhedron $Ax \leq b$.

We map

$$X \xrightarrow{(\mu, \sigma)} \mathbb{R}^2$$

each risky portfolio to the point $(\mu^\top x, \sigma(x))$ of the mean and variance of its return. In this article we disregard the entire distribution of the portfolio returns and just focus on those two statistics.

An important point on the mean-variance plane $\mathbb{R}^2$ is $(r_0, 0)$, which corresponds to riskless portfolios of deterministic return r_0 . We fix a $r_0 > 1$, so the rate by which a riskless asset's value grows in one time period.

The **Optimal Sharpe ratio problem** is

$$\max_{x \in X} S(x), \quad S(x) := \frac{\mu^\top x - r_0}{\sigma(x)}$$

here $S(x)$ is the Sharpe ratio for the portfolio x , which is the slope of the Capital Allocation Line through x .

Problem

$S(\cdot)$ is not a convex function, so the maximisation problem above is not convex. We are asked to derive a convex formulation for it.

We will do that by lifting the map S to a map $\hat{S}$ on a cone $\hat{X}$ generated by X , and then picking another slice of that cone, $X' \subset \hat{X}$, over which $\hat{S}$ is easily equivalent to a convex problem.

^{*}This article is written in the UK.

Construction of $\hat{X}$ and X'

Generate by $X \subset \mathbb{R}^n$ a cone in $\mathbb{R}^{n+1}$ by defining

$$\hat{X} := \{(px, p) \in \mathbb{R}^n \times \mathbb{R}_{\geq 0} : p > 0 \text{ and } x \in X\},$$

and extend

$$\hat{S}(px, p) := S(x) \tag{1}$$

(this is well-defined because each point in $\hat{X}$ is uniely expressible as (px, p)).

I imagine $\hat{X}$ as a multiplicatively expanding cone generated by X , where each point $x \in X$ corresponds to a ray in $\hat{X}$. If we denote the unit embedding $\epsilon(x) := (1 \cdot x, 1)$ and the projection $\pi(px, p) := x$, we get the following situation:

$$\begin{array}{ccc} X & \xrightleftharpoons[\pi]{\epsilon} & \hat{X} \\ & \searrow S & \downarrow \hat{S} \\ & & \mathbb{R} \end{array}$$

where all compositions with the same domain and ending in $\mathbb{R}$ trivially agree. The original problem optimizes on $\pi(\hat{X}) = X$, where S is not convex, and correspondingly $\hat{S}$ is not convex on $\epsilon(X) = \hat{X} \cap \{p = 1\}$. We will pick a different subset of $\hat{X}$ on which $\hat{S}$ trivially equivalent to a convex problem.

To that end, we observe that $\hat{S}$ can also be written as

$$\hat{S}(z, p) \equiv S(z/p) \equiv \frac{\mu^\top z/p - r_0}{\sigma(z/p)} \stackrel{!}{=} \frac{\mu^\top z - pr_0}{\sigma(z)}. \tag{2}$$

To eliminate the non-convexity here, we kill the numerator by picking that slice of $\hat{X}$ where the equality holds:

$$X' := \hat{X} \cap \{(z, p) : \mu^\top z - pr_0 = 1\}.$$

To let the abstract nonsense do the job, we have to show that X' is punctured uniquely by all rays in $\hat{X}$. This is obvious by considering the projection

$$\begin{aligned} \pi' : \hat{X} &\rightarrow X' \\ \pi'(px, p) &:= (p'_x x, p'_x), \\ \text{with } p'_x &:= 1/(\mu^\top x - r_0). \end{aligned}$$

The choice of the parameter p'_x ensures that $\pi' : \hat{X} \rightarrow X'$ has range X' and is a right inverse of the embedding $\epsilon' : X' \hookrightarrow \hat{X}$, $\epsilon'(px, p) := (px, p)$. The situation now is illustrated by:

$$\begin{array}{ccccc} X & \xrightleftharpoons[\pi]{\epsilon} & \hat{X} & \xrightleftharpoons[\pi']{\epsilon'} & X' \\ & & \downarrow \hat{S} & & \\ & & \mathbb{R} & \xleftarrow{\dots} & \end{array} \tag{3}$$

The maximum Sharpe ratio problem maximises the left vertical map here; we will now show that it is equivalent to maximise over the right one and how it enables a convex reformulation.

Convex reformulation

From the above discussion, the horizontal compositions in diagram (3)

$$X \xrightarrow{\pi' \circ \varepsilon} X' \text{ and } X' \xrightarrow{\pi \circ \varepsilon'} X$$

are inverses, so it is equivalent to optimize over X or X' . But over X' , $\hat{S}$ admits the simple expression

$$\hat{S}(z, p) \stackrel{(2)}{=} \frac{\mu^\top z - pr_0}{\sigma(z)} \stackrel{!}{=} \frac{1}{\sigma(z)}$$

because of the constraint $\mu^\top z - pr_0 = 1$, and so

$$\begin{aligned} \max_{x \in X} S(x) &= \max_{(z, p) \in X'} \hat{S}(z, p) = \max_{(z, p) \in X'} \frac{1}{\sigma(z)} \\ &= 1 / \min_{(z, p) \in X'} \sigma(z). \end{aligned}$$

Here $\sigma(z)$ is convex, and $(z, p) \in X'$ is a convex constraint because each constitutive constraint in

$$\begin{aligned} (z, p) \in X' &\Leftrightarrow z/p \in X \text{ and } \mu z - pr_0 = 1 \\ &\Leftrightarrow \forall i : f_i(z/p) \leq \tau_i \text{ and } Az/p \leq b \text{ and } 1 = \mu z - pr_0 \end{aligned}$$

is convex on its own:

1. $f_i(z/p) \leq \tau_i$ is convex because by f_i 's positive homogeneity it is equivalent to $f_i(z) \leq p\tau_i$, and so to $f_i(z) - p\tau_i \leq 0$, which is convex by convexity of f_i (part of the coherence condition assumed on each f_i);
2. $Az/p \leq b$ is just $Az \leq pb$, so linear and convex;
3. $1 = \mu z - pr_0$ is linear, and so convex, too.

It remains to recover the optimal x of $\max_x S(x)$ from the optimal (z, p) of $\min_{(z, p)} \sigma(z)$. By construction (1), for any $(z, p) \in \hat{X}$,

$$\hat{S}(z, p) = S(x), \quad \text{with } px = z,$$

so if a pair (z^*, p^*) maximises $\hat{S}(z, p)$, then $x^* := z^*/p^*$ maximises the Sharpe ratio S .

Thus we showed that the non-convex optimum Sharpe ratio problem over X is equivalent to a convex minimization one over X' by the relation

$$\begin{aligned} \max_{x \in X} S(x) &= 1 / \min_{(z, p) \in X'} \sigma(z). \\ x^* &\leftrightarrow (z^*, p^*) \\ \text{with } x^* &= z^*/p^*, \end{aligned}$$

where $x^* \leftrightarrow (z^*, p^*)$ is one-to-one because if $(z, p), (z', p') \in X'$ with $z/p = z'/p'$, then $z' = p'/p \cdot z$, substituted in the constraint $\mu^\top z' - p'r_0 = 1$, implies $\mu^\top z - pr_0 = p/p'$ forcing $p/p' = 1$, thus $p = p'$ and $z = z'$.